

Machine Learning in Failure Prediction in Breathalyzers

Ana Gleice Santos (✉ agsantos@inmetro.gov.br)

Instituto Nacional de Metrologia, Qualidade e Tecnologia

Luiz F. R. da C. Carmo

Instituto Nacional de Metrologia, Qualidade e Tecnologia

Charles B. Prado

Instituto Nacional de Metrologia, Qualidade e Tecnologia

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Abstract

To reduce accidents and deaths on roads, several countries have established laws that consider driving under the influence as a traffic violation [1, 2]. In Brazil, the instruments have to be in accordance with the OIML (R126) [3]. Errors in readings can result in unfair punishment if shows values above or below the real concentration. This study focused on identifying the probability of instrument failure before it occurs. This study addresses the use of classification models to predict failures that may occur between periodic checks performed every 12 months. The results showed the extra tree classifier model exhibited the best result, with an accuracy and precision of 75 and 77%, respectively. Machine learning models in metrological control is an alternative to add measurement reliability, providing legal certainty for the imposed infractions, and more accurately identifying drivers under the influence of alcohol and removing them from the roads.

Introduction

Many countries control traffic on public roads to prevent accidents and deaths caused by drunk drivers. Consequences attributed to drivers with alcohol levels higher than those allowed by law are generally serious, with high fines and lose your driver's license and your vehicle. Therefore, exact and precise measurement techniques were established to ensure that instruments indicated correct values, so as not to cause unfair punishment or the release of drunk drivers, for values above or below the true concentration, respectively. Control alternatives have been studied and addressed in studies ranging from the proposal to starting the engine only after the driver has passed alcohol tests with a touch-based system (on the surface of the skin) and the use of artificial intelligence to control traffic, using an image classification model [4,5,6].

Although new methodologies have been proposed, widely adopted control is still enforced through the use of breathalyzers to measure the mass concentration of alcohol in expired air [4]. Countries that have adopted this control evaluate models or accept models already approved by other countries that have adopted instrument evaluation criteria [7], thereby harmonizing the requirements for acceptance.

Instrument control steps involve the evaluation of the model, where the prototypes are tested in laboratories, simulating severe conditions of use, with variations in temperature, humidity, electromagnetic emissions, etc. Once the model is approved, each instrument must be submitted for initial verification and throughout its life by subsequent semi-annual or annual verifications [4].

The primary challenge for authorities is the establishment of safety criteria for imposing penalties on drivers. Verification indicates the measurement conditions at the time of testing. Once approved, there is no guarantee that the instrument will maintain its errors within limits until the next verification. Moreover, if an instrument fails, it is also impossible to determine the exact time at which the failure started, which is a challenge for legal metrology. In addition, this problem is encountered in other segments such as

public utility meters, where errors in reading consumption meters affect users and distributors [8]. Thus, incorrect results with breathalyzers can cause greater inconvenience.

One way to ensure that the driver is not unfairly penalized is to cancel all fines generated by an instrument, if the next subsequent verification produces a result that differs by more than the tolerance stated from the known value of the test standard. This control model was adopted by the United States [9]; however, it suffers from the disadvantage of not penalizing those who actually drive under the influence of alcohol.

Predicting instrument failures before they occur is essential to perform preventive actions. If a possible failure is detected in the metrological control of breathalyzers, the instrument can be sent for maintenance, or the period between checks can be reduced. This study proposed a method to evaluate the data generated in breathalyzer verifications, understand the main flaws found, and use machine learning classification models to label instruments in two categories: adequate and inadequate.

The remainder of this paper is organized as follows. Section 2 presents the main components in building a breathalyzer, while the recurring failures encountered during the life of the instrument are presented in Section 3. Next, certain examples in the literature on the application of fault detection using ML (machine learning) and an approach of how these alternatives can be used in metrology is presented. Further, similar studies are presented to exemplify the importance of this study, in Section 4. Section 5 presents the execution of data collection and pre-processing (Section 5), whereas the implementation of classification algorithms is presented in Section 6. Finally, the interpretation of data and suggestions for actions to be taken are provided in Section 7.

Breathalyzers And Their Construction

The breathalyzers used in traffic inspection follow technical performance requirements, with the aim of minimizing measurement errors when subjected to severe conditions of use, such as high and low temperatures and variations in relative humidity. The electromagnetic compatibility requirements are also assessed to establish that external emissions will not affect the measurement.

In terms of the technology employed, the primary ones are: infrared spectrometry (IR) and fuel cells. While the first is considered the most accurate, it is also the most expensive [10], which leads users to opt for the second on a larger scale. Brazil has approved models for both these technologies [11]; however, the instruments used are predominantly based on fuel cells.

The fuel cell construction scheme is briefly shown in Figure 1. The main components involved in the measurements can be clearly observed.

The measurement cycle begins by the sampling of a predetermined volume of exhaled air. The sensors control the temperature and flow, making necessary microprocessor-controlled adjustments. Subsequently, the alcohol present in the sample is electrically oxidized, releasing electrons. Thereafter, the

signals are converted into concentration values. The instrument does not directly measure the concentration of alcohol in the blood; rather it is measured through an existing physical relationship with exhaled air, which is approximately 2,100:1 (blood/exhaled air ratio) [10,12].

Main flaws found in breathalyzers

Knowing the construction of a breathalyzer, it can be inferred that failures in any of its main components can render measurements unfeasible or, in the worst case, cause incorrect readings.

The microprocessor monitors the entire measurement cycle, and the parameters are predetermined to enable this cycle. Further, faults in the temperature, flow, and pressure sensors are usually detected, and alarms are issued such that the user can repeat the operation and complete the measurement. However, the fuel cell generates a current caused by the oxidation of ethanol on its surface. As these cells age, or owing to other environmental factors [13], the process becomes less efficient, causing inaccurate readings that are not detected by the microprocessor. Therefore, this component is critical for the measurement process.

Data generated by repair and maintenance workshops (Figure 2) for breathalyzers used in Brazil showed that of the six main components, the fuel cell (or electrochemical) had the highest incidence of repair (40.2%) [14]. This was followed by the flow sensor (26.51%), which interrupts the reading if there is an interruption in the flow of the expired air. Next is the temperature sensor (16.96%), which ensures that the instrument does not operate outside the range specified by the manufacturer.

A temporal analysis revealed that, over the years, the incidence of repairs in fuel cells increased, becoming more evident in relation to the other components (Figure 3). The increase in fuel cell failures means an increase in the imprecision of the results generated by these instruments.

These failures imply legal uncertainties. They can serve as a basis for lawsuits brought against the state and cancel fines generated by instruments whose verification after the assessment failed. However, simply nullifying the generated results between checks is not the best way to solve this problem, as it may result in impunity for drivers who are actually driving under the influence of alcohol.

This scenario is most likely because errors found in verifications conducted by State Metrological Bodies [15] indicated negative values. In other words, when a reference solution was inserted to verify its accuracy, the instrument displayed lower values. (Figure 4). This suggests that the likelihood of a driver being unfairly fined is minimal. However, a large number of readings below the true value reflects the release of drivers who can cause traffic accidents due to driving under the influence of alcohol.

Thus, solutions that increase the reliability of metrological control of breathalyzers must be determined, as ignoring the occurrence of failures between checks can lead to unfair decisions, whereas, voiding all

punishments imposed on drivers can increase the occurrence of accidents, due to lack of credibility in the system.

Many studies have been conducted using machine learning models to predict instrument failures. Algorithms can understand behavior and detect anomalies and this trend has also been applied in metrology. The next section discusses examples of studies and implementations.

Machine Learning in Metrology

In metrology, the use of machine learning algorithms was reported by VIOREL-MIHAI [16] to identify the tendency of measurement instruments to fail. To create an algorithm, the authors used initial and subsequent verification data. The proposed methodology involved three steps: statistical control of compliance, establishment of control limits, and subsequent metrological verification. Consequently, the proposed model prevented instruments from presenting errors above the limits established in regulation until the next verification, thereby ensuring user reliability between the verifications performed. The conclusion of this study indicated the possibility of decision-making using the generated data.

Metrology faces many challenges in fulfilling its mission, such as metrological control of road scales. These instruments require the use of a standard weight corresponding to the usual bearing load, thereby rendering process exhausting, owing to the dimensions and transport conditions. To improve the control of these instruments, a study was conducted using a machine learning algorithm with neural networks [17]. Based on verification data collected over two years, the authors created a classification algorithm capable of identifying whether a given instrument would pass the indication error test, resulting in a reliability of 83.3%. Thus, knowing the advance trend of the result, users can take steps to reduce the weighing errors.

In the medical field, systems based on classifiers (Random Forest, Decision Tree, and KN-Neighbors) can reduce costs related to clinical engineering services and adapt maintenance protocols in defibrillators [18]. In this study, the authors used a set of measurement data acquired from periodic inspections, to overcome the challenges of maintenance protocols and increase clinical accuracy and efficiency in the diagnosis and treatment of patients.

In the area of instrument quality control, models based on regression and artificial neural networks (ANN) have been developed to predict measurement errors in instruments and standards [19]. The possibility of characterizing and quantifying measurement errors facilitates the performing of necessary corrections, scheduling the next calibration, or rejecting the instrument in advance. In addition to reducing measurement errors, the use of these algorithms provides savings for laboratories, as the general rule of thumb for a 12-month calibration interval can be extended.

With the increasing use of ML algorithms in measurement and analysis processes, their application in legal metrological control is a matter of time. Adapting the apps will be the next step. An important

example is the remote calibration of instruments [20]. Processes that combine intelligent instruments, communication technology, and machine learning can provide better remote calibration and reduce service time and cost.

A strong movement toward the application of artificial intelligence in the metrological control of instruments can be noticed with the cited examples. The possibility of using machine learning in all areas of legal metrology in Brazil is anticipated. In the next section, the application of this tool to the surveillance of breathalyzers used in traffic control is described.

Types of classification models

Machine-learning algorithms used for complex pattern recognition and process automation based on intelligent decisions are being used increasingly. This section highlights the regression and classification problems as subareas that are widely used in the construction of these algorithms.

Regression problems are defined as predictors to estimate numerical values in response; for example, a prediction of property value based on location, area, and the number of rooms and classification algorithms are responsible for predicting categories or classes based on the observation of behavior patterns, such as predicting whether a particular patient has a disease. The decision tree and linear classifier models are among the most commonly used methods.

The decision tree is similar to a flowchart. The most important input variable is considered to be the root node, and the other variables, as well as the labels, are considered as leaf nodes. The flow continues as functions run on each node. Following a pattern, it passes through all input variables and finds the best route to generate an answer (output). For these types of models, categorical data need not be converted to numeric data, which renders them a good choice for many classification problems.

An example of a decision tree model applied in metrology is the diagnosis of faults in wind turbine gears [21] and electronic equipment [22], as well as in the detection and classification of power quality disturbances [23]. Owing to their popularity, applications can be developed with several adaptations, including random forest and gradient-boosting algorithms that use clustering techniques to optimize the model.

Linear classifiers, classify data according to their characteristics within a space, through a line, plane, or hyperplane. Commonly used examples of these types of classifiers include Logistic Regression, Vector Machine Support, and Naive Bayes. In addition, in virtual metrology, they can be used for product quality control [24] and the detection and diagnosis of multiple failures in industrial processes [25].

In the implementation of this work, the results of certain classification techniques are compared, and according to the metrics, the best model for the problem addressed here is chosen.

Similar to the works cited above, this study presented a way to solve metrology gaps using machine learning techniques, defining trends in measurement instrument behavior, and increasing the reliability of the results.

Data collection and pre-processing

Integrating metrological control and artificial intelligence is a recent practice; however, existing databases were not created for this purpose, which renders it unfriendly and difficult to extract data for processing and analysis.

The two main metrological control data storage systems in Brazil: Integrated Management System (SGI) and Inmetro Services Portal in the States (PSIE). The first was conceived to standardize the processes and procedures of Inmetro's Brazilian Metrology and Quality Network (RBMLQ-I) [26]. Whereas, the second was built to increase the reliability of the repaired measuring instruments because this service is performed by private entities [27].

For both, access is restricted to authorized persons; therefore, an external user cannot obtain such information. Once the locations for data extraction were identified, data were collected. Considering that each breathalyzer was checked, on average, every 12 months, a relatively long period to capture information about the instrument's behavior and its drift must be established. Thus, the period studied comprised five years (2016–2020).

Sampling and preprocessing are decisive steps for the quality of the analysis. The raw data collected included both categorical and numerical variables (Table 1). Among the categories, information on the types of models, reference materials, laboratories responsible for tests, and others was available. Further, for numerical variables, data on the concentrations of the reference solutions, test dates, errors, and standard deviations were found. In addition to these, the 'drift' variable was proposed, as the aim was to explicitly understand the error variation over the years.

The selection of predictor variables is an important step in building a model. Many analysts use feature selection techniques to find the best variables, thus reducing computation time and improving the algorithm performance [28].

The goal was to predict instrument failures before their occurrence, and it is known that these failures are directly related to fuel-cell wear. The following predictive variables were selected: error indications, standard deviation, and deviation for each concentration (Table 2), as they can provide indications of instrument failure. In addition, to reduce noise and standardize the types of inputs, only samples whose tests were performed with reference materials in gas mixtures and performed by a single laboratory were considered. Consequently, although the generalization power of the algorithm was reduced, the reproducibility of the technique was improved. Because this was a supervised learning classification

problem, samples were separated and labeled as suitable "0" and inappropriate "1" if they passed or failed the next check, respectively.

Table 1 - Information available in the integrated management system for breathalyzers

Categorical variables	Numerical variables
Service description	Date
Customer	Quantity by month, year, State and region
Type of test method (liquid or gas)	Total planned and executed
Serial number	Financial by Region and State
Result (passed, failed)	Results (error and standard deviation)
Brand and model	
Test type (initial or subsequent)	
verification mark	

Table 2 - Input variables used for breathalyzer classification models

Input Variables		
Mean Error conc. I	Mean Error conc. II	Mean Error conc. III
Deviation conc. I	Deviation conc. II	Deviation conc. III
Drift I	Drift II	Drift III

A common problem when using supervised learning in classification algorithms is finding unbalanced data. The same was observed for this problem; the samples selected for training had an approximate ratio of 1:6 (samples 1 and 0, respectively) (Figure 5). Training the model with unbalanced data can result in more learning of the characteristics of the majority class, while the minority class may often be misclassified [29]. Among the various existing techniques to eliminate the disproportion of class types, this study combined the Neighbor Cleaning Rule (NLC) and Random Under Sample (RUS), which are responsible for removing outliers and excluding samples from the majority class, respectively. Once the preprocessing phase was concluded, the models were applied, and their metrics were analyzed

Implementation

In this section, the machine-learning algorithm that best fits the training data is discussed. Currently, dozens of models and model adaptations are used in classification problems. As the focus of this study is not related to the construction of codes, rather their application to improve metrological control, auto-ML was considered to investigate the most varied models and quickly determine the one that best explained and predicted possible failures in breathalyzers before they occur.

Auto-ML is the result of automating machine-learning models. Automation considers that repetitive processes of implementation, such as data standardization, hyperparameter adjustments, training and test set division, evaluation, and validation, can be optimized without the need for manual configurations [30]. Further, it accelerates the cycle of experiments, thereby making processes more productive.

After preprocessing the dataset, the PyCaret library (an open-source Auto-ML in Python) was used, with samples divided as 70% for training and 30% for testing.

PyCaret ranks the trained models according to the best performance metrics (Table 3). Further, the precision metric was applied by default for the classification problems. With the input variables, the Extra Tree Classifier model was used as the best classification.

The proposed approach attempts to observe the results and also propose actions. The statistics that the model returns to the labels must be known. For this, the confusion matrix with the Extra Tree Classifier model was plotted (Figure 6) to visualize the distribution of the true positive, false positive, true negative, and false negative results.

Table 3. Ranking models ordered by PyCaret according to the performance of its metrics

Model	Accuracy	AUC	Recall	Prec.	F1	Kappa
Extra Trees Classifier	0.7556	0.8069	0.7447	0.7717	0.7530	0.5116
Gradient Boosting Classifier	0.7178	0.7823	0.6960	0.7348	0.7135	0.4360
Light Gradient Boosting Machine	0.7178	0.7731	0.7185	0.7293	0.7199	0.4361
Random Forest Classifier	0.7156	0.7958	0.7181	0.7231	0.7183	0.4315
Ada Boost Classifier	0.6978	0.7646	0.6738	0.7199	0.6926	0.3960
Linear Discriminant Analysis	0.6844	0.7408	0.5984	0.7299	0.6530	0.3693
Naive Bayes	0.6778	0.7494	0.5018	0.7803	0.6077	0.3573
Quadratic Discriminant Analysis	0.6644	0.7446	0.5108	0.7479	0.6031	0.3305
Ridge Classifier	0.6533	0.0000	0.5501	0.6976	0.6112	0.3076
K Neighbors Classifier	0.6511	0.7366	0.5814	0.6813	0.6254	0.3032
Logistic Regression	0.6400	0.7272	0.5455	0.6731	0.6001	0.2808
Decision Tree Classifier	0.6311	0.6311	0.6300	0.6350	0.6317	0.2621
SVM - Linear Kernel	0.6267	0.0000	0.5725	0.6689	0.5512	0.2544
Dummy Classifier	0.5044	0.5000	10.000	0.5044	0.6706	0.0000

The proposed approach attempts to observe the results and also propose actions. The statistics that the model returns to the labels must be known. For this, the confusion matrix with the Extra Tree Classifier model was plotted (Figure 6) to visualize the distribution of the true positive, false positive, true negative, and false negative results.

To graphically visualize the performance of the model, the Receptor operator characteristic (ROC) curve that returns the rate of true positives (sensitivity) on the y-axis against the rate of false positives (1 - specificity) on the x-axis is shown (Figure 7)

Here, a similar performance for classes 0 and 1 was observed. The area under the ROC curve (0.77) was considered acceptable discrimination [31]. When considering this metric, primarily, the costs required when the model identified a false-positive result was assessed. That is, the implications caused when breathalyzers are identified as fit until the next check but fail at certain point during their 12-month validity.

Assessing the unnecessary costs that the state would incur is a way of optimizing public spending. Thus, the scenario must be known, whose metrics can identify false negatives; that is, instruments labeled as inadequate, which will in fact maintain their metrological characteristics within established standards.

The false negative rate is also critical in classifier analysis. In this case, this would imply instruments labelled as adequate, which will in fact fail at a certain point until the next check.

Upon obtaining the results of the model, decision makers can direct their efforts toward actions that have a positive impact on society. In the next section, the interpretation of the results obtained is discussed.

Interpretation And Conclusions

Metrology is committed to ensuring accurate measurements. Therefore, alternatives must be developed to improve the processes inherent to the instruments and measurement methods. In this work, we use classification algorithms to predict failures in breathalyzers used in traffic inspections.

The results showed the complexity of understanding the behavior of breathalyzers in use before they present an error above the allowed limit, causing inaccurate measurements in the field that impact road safety or the imputation of unfair punishments.

The metrics confirmed the extra tree classifier model as the best classifier, with accuracy and precision of 75 and 77%, respectively. While satisfying, it leaves the task of managing false-negative and false-positive results to the authorities.

In the development of this study, it was necessary to analyze the data before applying the models, to understand their distribution, constructive components and regularity in the presentation of defects. The combination of computer language, statistical analysis and metrological control will provide a great advance for metrology.

The main objective of this study was achieved; however, new issues were identified, which will give rise to future work. The first is concerned with improving the architecture of the databases to facilitate extraction and processing.

The second issue relates to enabling access levels for external users. In addition to providing transparency to metrological control, it will allow other researchers and analysts to conduct studies and suggest improvements. In parallel with this availability, information security must be considered to ensure the privacy of the instrument holders.

Finally, this study suggested the use of machine learning in the construction of digital twins, virtually representing instruments and measurement processes. Through this, the metrological control can be systemized using concrete applications of artificial intelligence. This mode of action will only be possible with the training of metrologists and investment in infrastructure and IT security. Further, machine learning provides lower costs and higher productivity and quality, which results in greater confidence in the measurement process.

References

1. LI Y, SZE N, WONG S, TSUI K, SO F. Effects of Drink Driving on Crash Risk Based on Random Breath Test Data. *Hong Kong Journal of Emergency Medicine*. 2013;20(3):146-154. doi:10.1177/102490791302000304. 2013
2. ALAM GONÇALVES GUIMARÃES, ALAN RICARDO DA SILVA. Impact of regulations to control alcohol consumption by drivers: An assessment of reduction in fatal traffic accident numbers in the Federal District, Brazil, *Accident Analysis & Prevention*, Volume 127, Pages 110-117, ISSN 0001-4575, <https://doi.org/10.1016/j.aap.2019.01.017>. 2019
3. OIML Recommendation R 126 Evidential Breath Analyzers. 2021
4. MANDAL, NITESH et al. Vehicle Tracking with Alcohol Detection & Seat Belt Control System. In: 2020 International Conference for Emerging Technology (INCET). IEEE, p. 1-5. 2020.
5. JONES, T.; PRISCILLA, A.; RUTH, E. Intelligent Safety and Security Systems in Automobiles. Special Issue Published in *International Journal of Trend in Research and Development (IJTRD)*, ISSN: 2394-9333, www.ijtrd.com. September 2017.
6. WANZALA, J. N.; ATIM, M. R. AI Based Wireless Traffic Light Detector using Image Classification in Machine Learning to Reduce Road Traffic Accidents: A Case Study of Uganda's Road Traffic. 2021.
7. Ehrlich, C. Considerations and Concepts in the Development of the Framework for Mutual Acceptance Arrangement on OIML Type Evaluations, *Asia-Pacific Legal Metrology Forum Circular*, [online], https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=904076. 2004.
8. PAŁASZ, PRZEMYSŁAW; PRZYSOWA, RADOSŁAW. Using Different ML Algorithms and Hyperparameter Optimization to Predict Heat Meters' Failures. *Applied Sciences*, v. 9, n. 18, p. 3719, 2019.

9. Code of Federal Regulations, Subpart E, "§ 26.91 Acceptable devices for conducting initial and confirmatory tests for alcohol and methods of use." 57 FR 55072, Nov. 24, 1992; 73 FR 17197 Mar. 31, 2008.
10. OZOEMENA, Kenneth I. et al. Fuel cell-based breath-alcohol sensors: Innovation-hungry old electrochemistry. *Current Opinion in Electrochemistry*, v. 10, p. 82-87, 2018.
11. BRAZIL. Ministry of Industry, Foreign Trade and Services. INMETRO. Bases Legislativas (pesquisa avançada). <<http://www.inmetro.gov.br/legislacao>>
12. HARDING, Patrick; FIELD, Patricia H. Breathalyzer® accuracy in actual law enforcement practice: A comparison of blood-and breath-alcohol results in Wisconsin drivers. *Journal of Forensic Science*, v. 32, n. 5, p. 1235-1240, 1987.
13. PREST, LAURA. Fundamental investigation of fuel cell-based breath alcohol sensors and the cause of sensor degradation in low-humidity conditions. 2011. URI: <http://hdl.handle.net/10155/204>.
14. PSIE - Portal de Serviços do Inmetro nos Estados. Inmetro, accessed October – December 2020, <<https://servicos.rbmlq.gov.br>>
15. SGI - Sistema de Gestão Integrada. Inmetro, accessed April – December 2020, <https://sgi.inmetro.rs.gov.br/>
16. VIOREL-MIHAI, NANI. A new algorithm for predictive metrological verification of measuring apparatus. *Journal: International Journal of Precision Engineering and Manufacturing - Volume 19, Issue 2*, pp. 167-172 – 2018
17. KLIMENT, Tomáš et al. Diagnosis of the Accuracy of the Vehicle Scale Using Neural Network. *Measurement Science Review*, v. 19, n. 1, p. 14-19, 2019.
18. BADNJEVIĆ, ALMIR; POKVIĆ, LEJLA GURBETA; HASIČIĆ, MEHRIJA; BANDIĆ, LEJLA; MAŠETIĆ, ZERINA; KOVAČEVIĆ, ŽIVORAD; KEVRIĆ, JASMIN; PECCHIA, LEANDRO. Evidence-based clinical engineering: Machine learning algorithms for prediction of defibrillator performance. *Biomedical Signal Processing and Control*, Volume 54, 2019, 101629, ISSN 1746-8094.
19. VINODKUMAR JACOB. Models for Prediction of Measurement Errors using Regression Analysis and Artificial Neural Networks. *Turkish Journal of Computer and Mathematics Education* Vol.12 No.10 (2021), 2552-2558.
20. WANG, QUAN, HONGBIN LI, HAO WANG, JUN ZHANG, AND JILIANG FU. 2022. "A Remote Calibration Device Using Edge Intelligence". *Sensors* 22, no. 1: 322. <https://doi.org/10.3390/s22010322>.
21. PANG B, TIAN T, TANG G-J. Fault state recognition of wind turbine gearbox based on generalized multi-scale dynamic time warping. *Structural Health Monitoring*. 2021; 20(6): 3007-3023. doi:10.1177/1475921720978622.
22. R. VAEZ-GHAEMI, W GODBERSEN, H SCHWETLICK, D FILBERT. Some aspects of knowledge-based fault diagnosis in electronic devices. *Measurement*, Volume 10, Issue 1, Pages 2-7, ISSN 0263-2241, [https://doi.org/10.1016/0263-2241\(92\)90033-Z](https://doi.org/10.1016/0263-2241(92)90033-Z). 1992
23. RIBEIRO, EDUARDO G.; MENDES, THAIS M.; DIAS, GUILHERME L.; FARIA, EULER R.S. VIANA, FELIPE M.; BARBOSA, BRUNO H.G.; FERREIRA, DANTON D.; Real-time system for automatic detection and

- classification of single and multiple power quality disturbances. *Measurement*, Volume 128, Pages 276-283, ISSN 0263-2241, <https://doi.org/10.1016/j.measurement.2018.06.059>. 2018
24. TILOUCHE, S.; BASSETTO, S. AND NIA, V. P. "Classification algorithms for virtual metrology," *2014 IEEE International Conference on Management of Innovation and Technology*, pp. 495-499, doi: 10.1109/ICMIT.2014.6942477. 2014
 25. CHEN JING, JIAN HOU. SVM and PCA based fault classification approaches for complicated industrial process. *Neurocomputing*, Volume 167, Pages 636-642, ISSN 0925-2312, <https://doi.org/10.1016/j.neucom.2015.03.082>. 2015
 26. SILVA, JANARA MARIA VEIGA SILVA. Estudo do Sistema de Gestão Integrada (SGI) como fator preponderante para a tomada de decisão na Rede Brasileira de Metrologia Legal, e Qualidade Inmetro (RBMLQ-I) na área da Fiscalização da Qualidade. 2013. 150 f. Trabalho de Conclusão de Curso de Mestrado em Metrologia e Qualidade – Instituto Nacional de Metrologia, Qualidade e Tecnologia, 2013.
 27. RODRIGUES FILHO, B. A.; SILVA, M. E. The PSIE platform as a tool to improve metrological control in Brazil. In: *4th CAFMET Proceedings*. p. 3. 2014
 28. CHANDRASHEKAR, GIRISH; SAHIN, FERAT. A survey on feature selection methods. *Computers & Electrical Engineering*, Volume 40, Issue 1, Pages 16-28, ISSN 0045-7906, <https://doi.org/10.1016/j.compeleceng.2013.11.024>. 2014.
 29. KHOSHGOFTAAR, T. M.; GAO, K. AND SELIYA, N., "Attribute Selection and Imbalanced Data: Problems in Software Defect Prediction," *2010 22nd IEEE International Conference on Tools with Artificial Intelligence*, pp. 137-144, doi: 10.1109/ICTAI.2010.27. 2010
 30. PRASAD, D. VENKATA VARA; KUMAR, P. SENTHIL; VENKATARAMANA, LOKESWARI Y.; PRASANNAMEHA, G.; HARSHANA, S.; SRIVIDYA, S. JAHNAVI; HARRINEI, K.; INDRAGANTI, SRAVYA. Automating water quality analysis using ML and auto ML techniques. *Environmental Research*, Volume 202, 111720, ISSN 0013-9351. 2021.
 31. HOSMER, D.W. AND LEMESHOW, S. *Applied Logistic Regression*. - 2nd ed. John Wiley & Sons, Inc. Pp. 156-164, 2000.

Figures

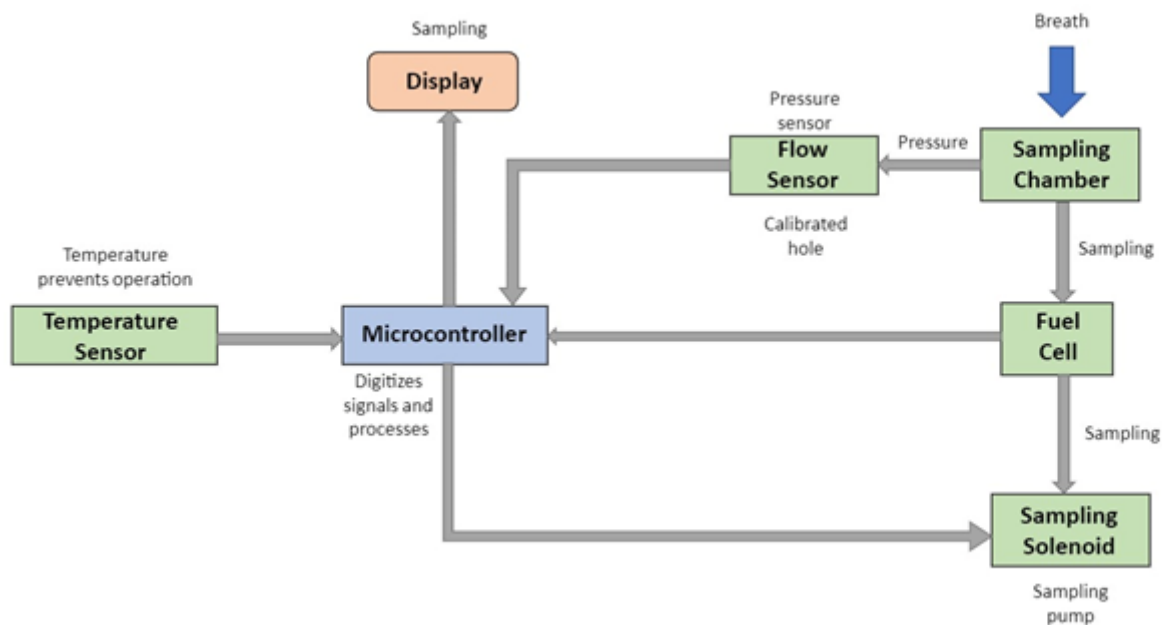


Figure 1

Diagram of the measurement cycle of breathalyzers developed with a fuel cell.

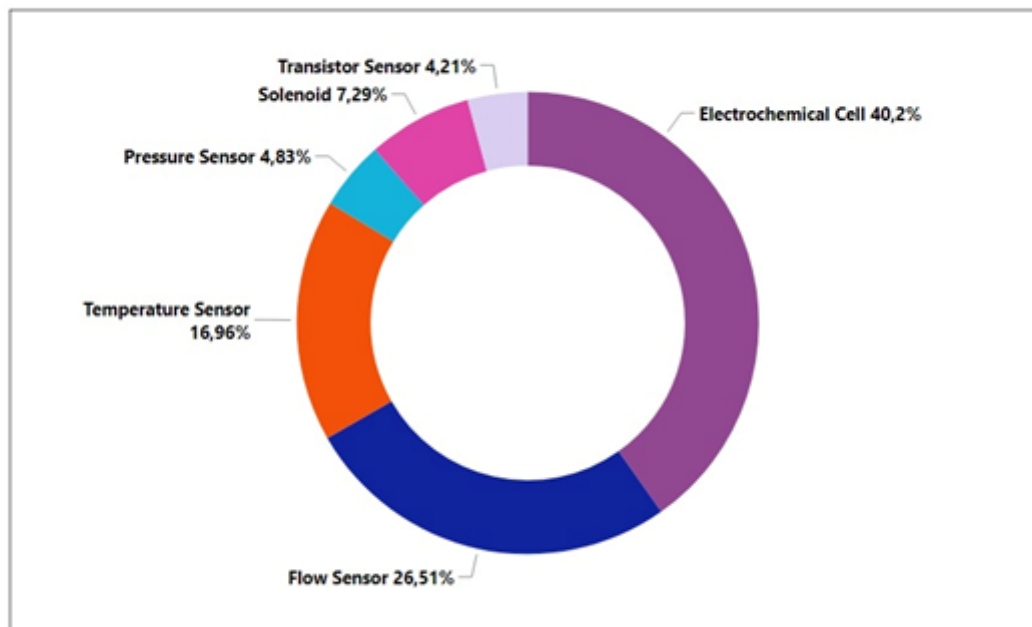


Figure 2

Distribution of frequency of repairs performed on breathalyzers by authorized workshops in Brazil (data from 2017 to 2020)

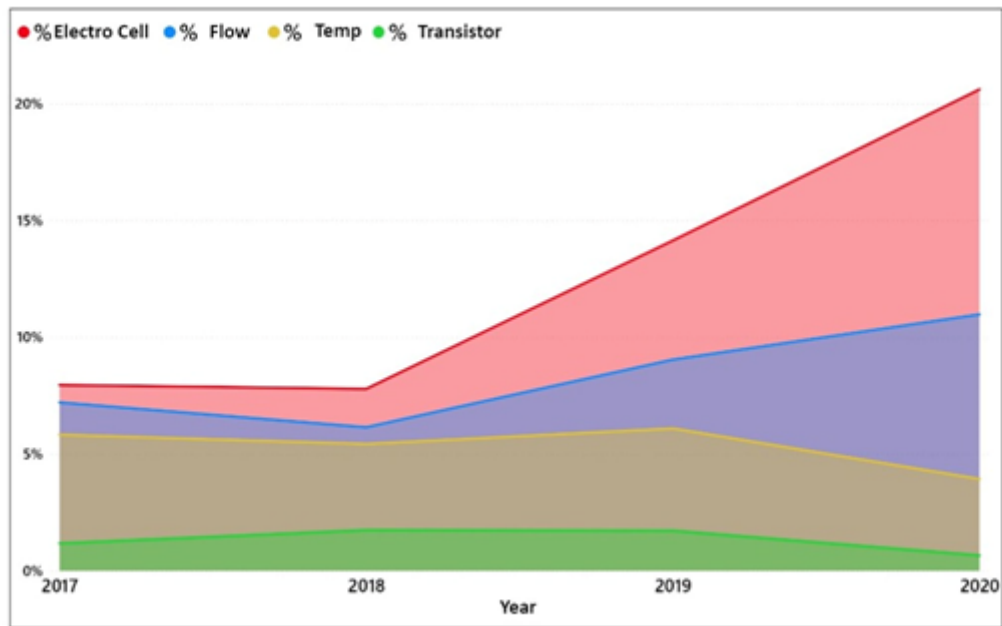


Figure 3

Area chart of the cumulative total of repairs to the breathalyzer components in the years 2017 to 2020.

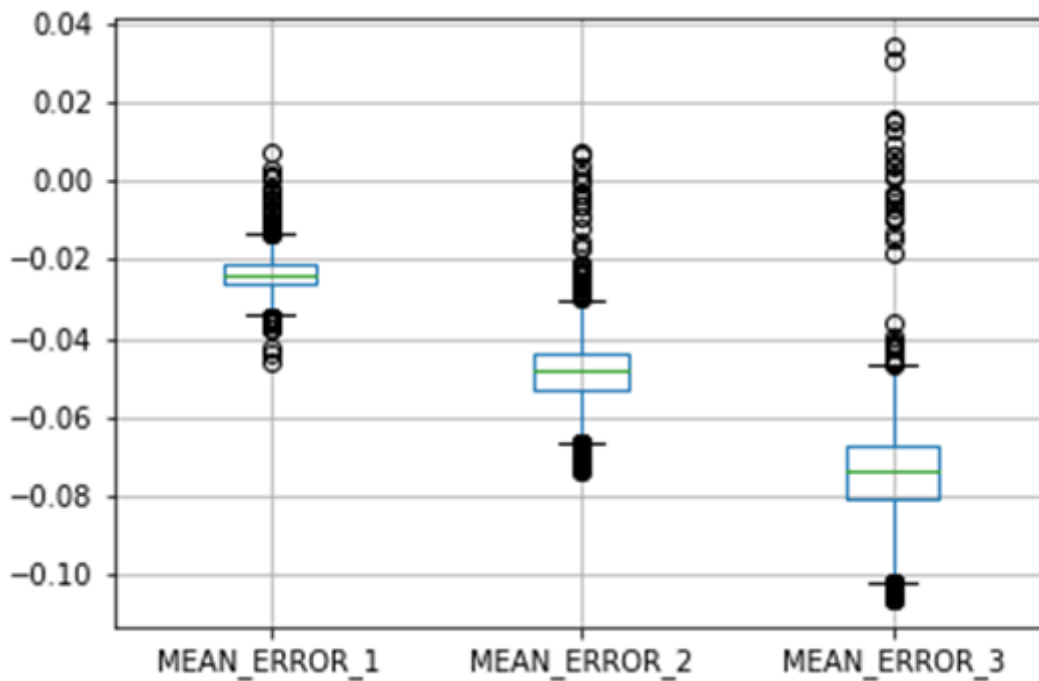


Figure 4

Distribution of errors found during breathalyzer verification, showing an asymmetry, with a predominance of negative values.

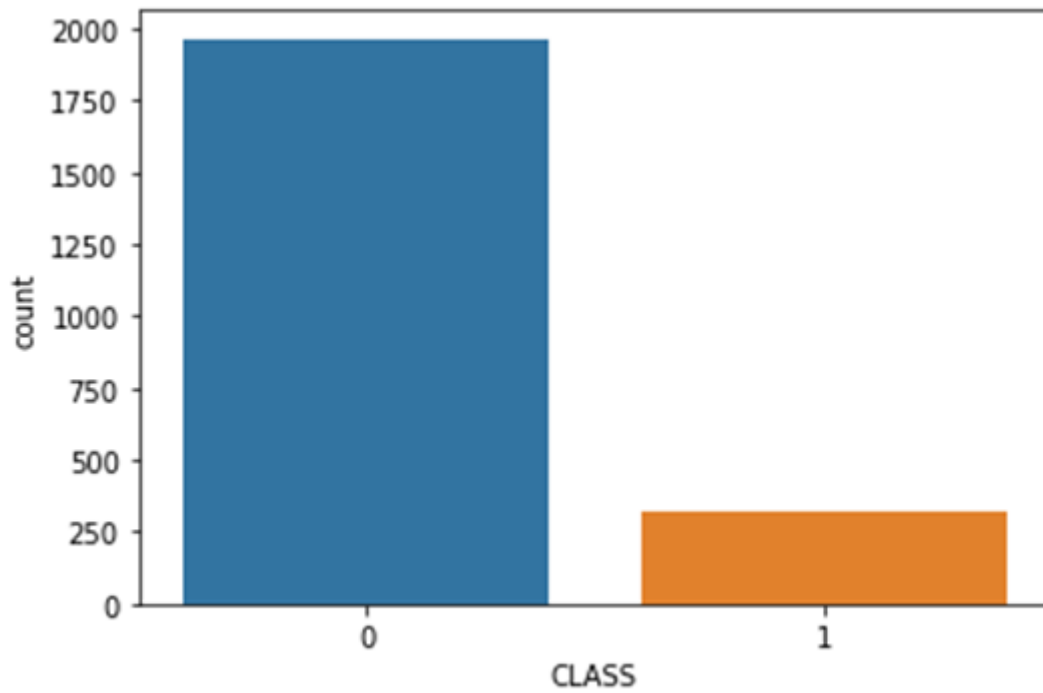


Figure 5

Class count: 0 for instruments that did not fail subsequent verification and 0 for those that failed

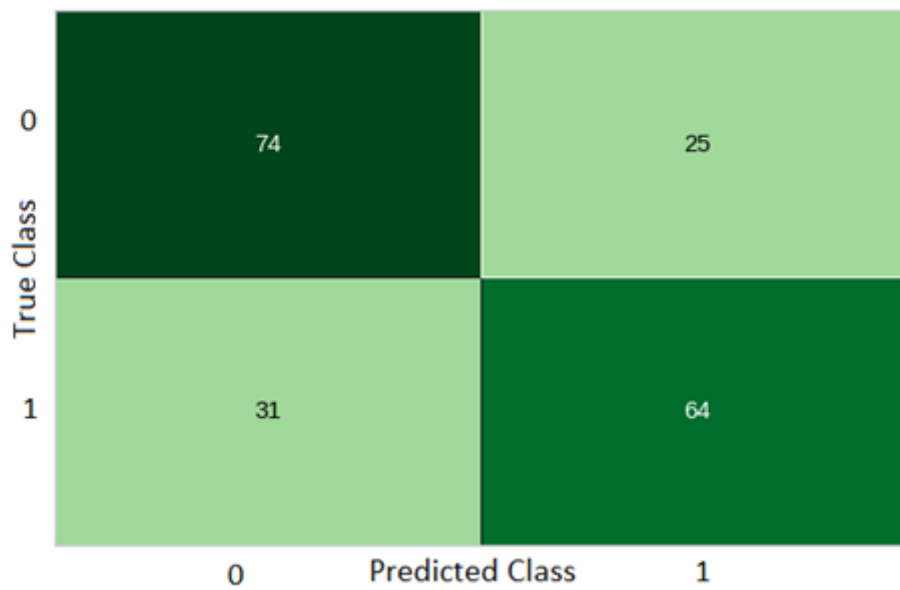


Figure 6

Confusion matrix for Extra Tree Classifier model

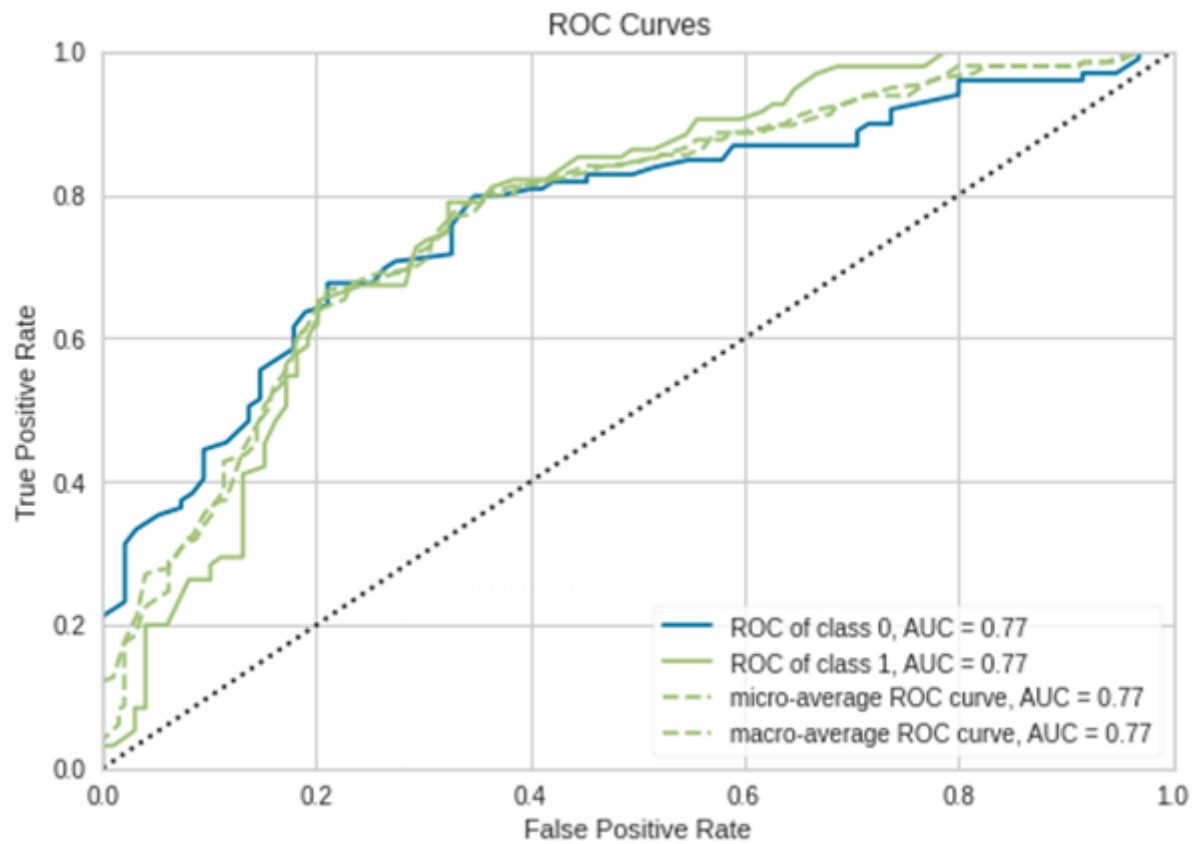


Figure 7

ROC curve for the Extra Trees Classifier model, result of prediction for classes 0 and 1, both with 77% accuracy